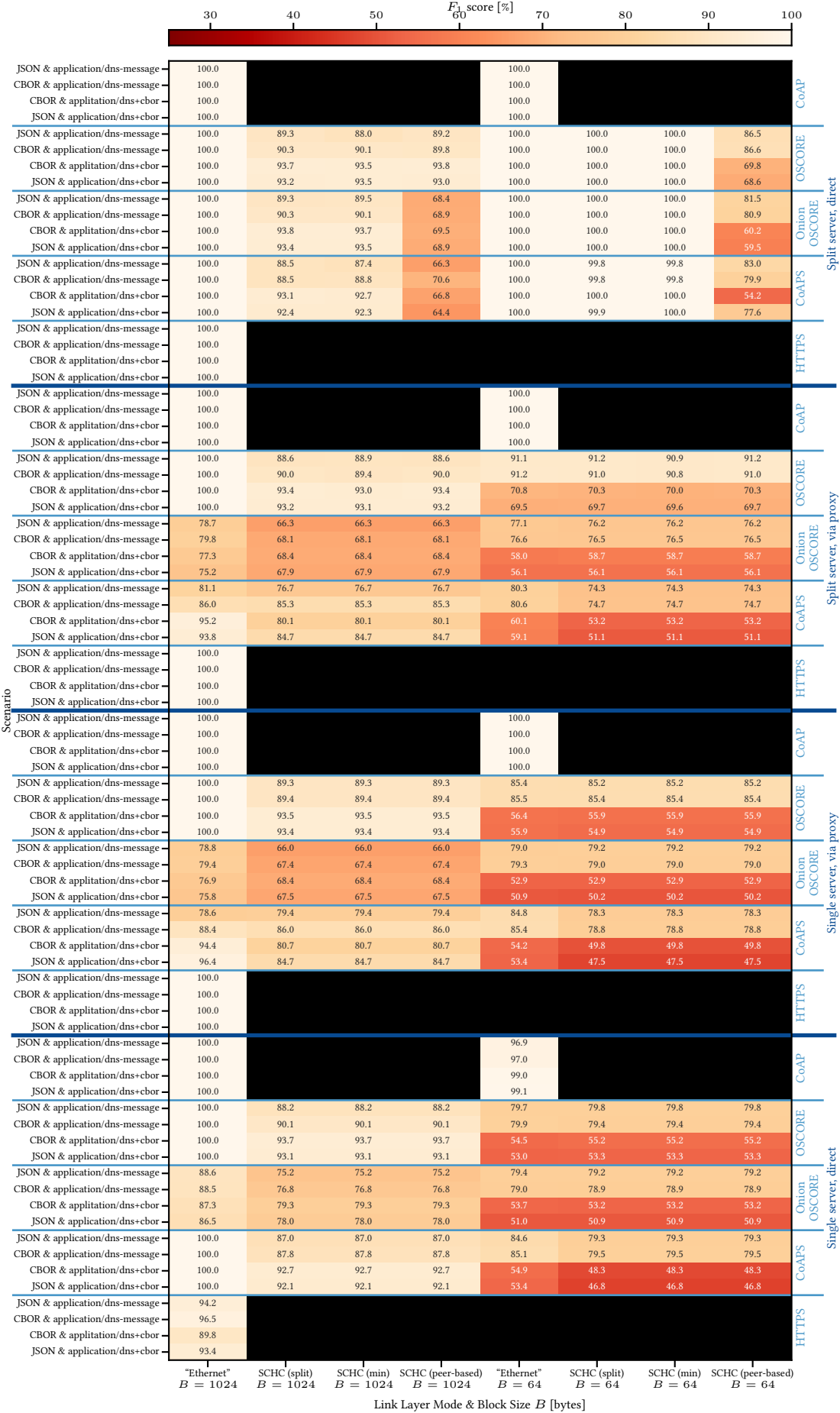
$F_1$ score [%]
30
40
50
60
70
80
90
100
Scenario
Link Layer Mode & Block Size $B$ [bytes]
"Ethernet" $B = 1024$
SCHC (split) $B = 1024$
SCHC (min) $B = 1024$
SCHC (peer-based) $B = 1024$
"Ethernet" $B = 64$
SCHC (split) $B = 64$
SCHC (min) $B = 64$
SCHC (peer-based) $B = 64$
Split server, direct
CoAP
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
OSCORE
JSON & application/dns-message 100.0 89.3 88.0 89.2 100.0 100.0 100.0 86.5
CBOR & application/dns-message 100.0 90.3 90.1 89.8 100.0 100.0 100.0 86.6
CBOR & applitation/dns+cbor 100.0 93.7 93.5 93.8 100.0 100.0 100.0 69.8
JSON & application/dns+cbor 100.0 93.2 93.5 93.0 100.0 100.0 100.0 68.6
Onion OSCORE
JSON & application/dns-message 100.0 89.3 89.5 68.4 100.0 100.0 100.0 81.5
CBOR & application/dns-message 100.0 90.3 90.1 68.9 100.0 100.0 100.0 80.9
CBOR & applitation/dns+cbor 100.0 93.8 93.7 69.5 100.0 100.0 100.0 60.2
JSON & application/dns+cbor 100.0 93.4 93.5 68.9 100.0 100.0 100.0 59.5
CoAPS
JSON & application/dns-message 100.0 88.5 87.4 66.3 100.0 99.8 99.8 83.0
CBOR & application/dns-message 100.0 88.5 88.8 70.6 100.0 99.8 99.8 79.9
CBOR & applitation/dns+cbor 100.0 93.1 92.7 66.8 100.0 100.0 100.0 54.2
JSON & application/dns+cbor 100.0 92.4 92.3 64.4 100.0 99.9 100.0 77.6
HTTPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 100.0
Split server, via proxy
CoAP
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
OSCORE
JSON & application/dns-message 100.0 88.6 88.9 88.6 91.1 91.2 90.9 91.2
CBOR & application/dns-message 100.0 90.0 89.4 90.0 91.2 91.0 90.8 91.0
CBOR & applitation/dns+cbor 100.0 93.4 93.0 93.4 70.8 70.3 70.0 70.3
JSON & application/dns+cbor 100.0 93.2 93.1 93.2 69.5 69.7 69.6 69.7
Onion OSCORE
JSON & application/dns-message 78.7 66.3 66.3 66.3 77.1 76.2 76.2 76.2
CBOR & application/dns-message 79.8 68.1 68.1 68.1 76.6 76.5 76.5 76.5
CBOR & applitation/dns+cbor 77.3 68.4 68.4 68.4 58.0 58.7 58.7 58.7
JSON & application/dns+cbor 75.2 67.9 67.9 67.9 56.1 56.1 56.1 56.1
CoAPS
JSON & application/dns-message 81.1 76.7 76.7 76.7 80.3 74.3 74.3 74.3
CBOR & application/dns-message 86.0 85.3 85.3 85.3 80.6 74.7 74.7 74.7
CBOR & applitation/dns+cbor 95.2 80.1 80.1 80.1 60.1 53.2 53.2 53.2
JSON & application/dns+cbor 93.8 84.7 84.7 84.7 59.1 51.1 51.1 51.1
HTTPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 100.0
Single server, via proxy
CoAP
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
OSCORE
JSON & application/dns-message 100.0 89.3 89.3 89.3 85.4 85.2 85.2 85.2
CBOR & application/dns-message 100.0 89.4 89.4 89.4 85.5 85.4 85.4 85.4
CBOR & applitation/dns+cbor 100.0 93.5 93.5 93.5 56.4 55.9 55.9 55.9
JSON & application/dns+cbor 100.0 93.4 93.4 93.4 55.9 54.9 54.9 54.9
Onion OSCORE
JSON & application/dns-message 78.8 66.0 66.0 66.0 79.0 79.2 79.2 79.2
CBOR & application/dns-message 79.4 67.4 67.4 67.4 79.3 79.0 79.0 79.0
CBOR & applitation/dns+cbor 76.9 68.4 68.4 68.4 52.9 52.9 52.9 52.9
JSON & application/dns+cbor 75.8 67.5 67.5 67.5 50.9 50.2 50.2 50.2
CoAPS
JSON & application/dns-message 78.6 79.4 79.4 79.4 84.8 78.3 78.3 78.3
CBOR & application/dns-message 88.4 86.0 86.0 86.0 85.4 78.8 78.8 78.8
CBOR & applitation/dns+cbor 94.4 80.7 80.7 80.7 54.2 49.8 49.8 49.8
JSON & application/dns+cbor 96.4 84.7 84.7 84.7 53.4 47.5 47.5 47.5
HTTPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 100.0
Single server, direct
CoAP
JSON & application/dns-message 100.0 96.9
CBOR & application/dns-message 100.0 97.0
CBOR & applitation/dns+cbor 100.0 99.0
JSON & application/dns+cbor 100.0 99.1
OSCORE
JSON & application/dns-message 100.0 88.2 88.2 88.2 79.7 79.8 79.8 79.8
CBOR & application/dns-message 100.0 90.1 90.1 90.1 79.9 79.4 79.4 79.4
CBOR & applitation/dns+cbor 100.0 93.7 93.7 93.7 54.5 55.2 55.2 55.2
JSON & application/dns+cbor 100.0 93.1 93.1 93.1 53.0 53.3 53.3 53.3
Onion OSCORE
JSON & application/dns-message 88.6 75.2 75.2 75.2 79.4 79.2 79.2 79.2
CBOR & application/dns-message 88.5 76.8 76.8 76.8 79.0 78.9 78.9 78.9
CBOR & applitation/dns+cbor 87.3 79.3 79.3 79.3 53.7 53.2 53.2 53.2
JSON & application/dns+cbor 86.5 78.0 78.0 78.0 51.0 50.9 50.9 50.9
CoAPS
JSON & application/dns-message 100.0 87.0 87.0 87.0 84.6 79.3 79.3 79.3
CBOR & application/dns-message 100.0 87.8 87.8 87.8 85.1 79.5 79.5 79.5
CBOR & applitation/dns+cbor 100.0 92.7 92.7 92.7 54.9 48.3 48.3 48.3
JSON & application/dns+cbor 100.0 92.1 92.1 92.1 53.4 46.8 46.8 46.8
HTTPS
JSON & application/dns-message 94.2
CBOR & application/dns-message 96.5
CBOR & applitation/dns+cbor 89.8
JSON & application/dns+cbor 93.4